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Web and Mobile Application Development

Week 11

Comparative Analysis Table

Category	Native (Android/iOS)	React Native	Flutter	
Programming Language	Kotlin/Java (Android), Swift (iOS)	JavaScript / TypeScript	Dart	
UI Components	Real native UI components	Native components via bridge	Own widgets (custom UI)	
Performance	Best performance	Good, slightly slower	Very high, close to native	
Learning Curve	Hard (2 platforms)	Easy (web dev friendly)	Medium (learn Dart)	
Community Support	Very strong	Very large	Growing fast (Google support)	

After comparing Native development, React Native, and Flutter, the most compelling hybrid framework is Flutter. This is mainly because of its strong performance, consistent user interface, and modern development approach. Flutter stands out because it does not rely on a bridge like React Native. Instead, it compiles directly into native code, which improves speed and reduces performance issues. This allows Flutter apps to run smoothly, even with complex animations or graphics. Another advantage of Flutter is its UI system. Unlike React Native, which depends on native components, Flutter uses its own widgets. This ensures that the app looks the same on both Android and iOS. Although React Native is easier to learn because it uses JavaScript, Flutter provides a more complete and controlled development environment. Its performance is often closer to native apps while still using a single codebase. Overall, Flutter is the best choice for developers who want high performance, better UI control, and cross-platform efficiency.

“Hello World” Tutorial (Flutter)

Steps Followed:

Installed Flutter SDK

Installed Android Studio

Set up an Android emulator

Created a new Flutter project using: flutter create hello_app

Opened the project in Android Studio

Located the main.dart file

Modified the code to display Hello World

Ran the app using: flutter run

The emulator launched and displayed the app

Screenshots (Add in Word)

Screenshot 1: Flutter project created
Screenshot 2: Code showing Hello World
Screenshot 3: Emulator running the app

Reflection

Using Flutter to build a simple Hello World app was a different experience compared to native Android development. One major difference is how fast and simple the setup process is once everything is installed. Flutter provides a ready-made structure, which makes it easy to start building apps quickly. Compared to native Android development using Java or Kotlin, Flutter feels easier because you only write one codebase for both Android and iOS. In native development, you would need separate code for each platform, which takes more time and effort. Flutter also has a feature called hot reload, which allows you to see changes instantly without restarting the app. This makes development faster and more efficient. However, learning Dart was slightly challenging at first since it is a new language. Overall, Flutter makes mobile development faster, easier, and more flexible compared to traditional methods.